A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft Trajectory

ISD 503 Capstone Project Report – Interim Report #1

Submitted to:
Mark Petrotta - Principal Engineer of Digital Engineering, MBSE, and Augmented Intelligence - Aerospace and Defense Systems
IOE Department of Integrated Systems Engineering & Design

Prepared by:
Owen O'Keefe
University of Michigan, ISD

October 10, 2026

Plan ref: to-do-list.md S1 — Confirm advisor/reader name and title for "Submitted to."

Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 5.0 Impact/Financial Benefits), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report.

Contents

Executive Summary **1**

1 Introduction and Current State **2**
- 1.1 Overview of the Project . . . 3
- 1.2 Review of Current State Design / Literature Review . . . 4
- 1.3 Stakeholders . . . 5
- 1.4 Scope . . . 7
- 1.5 Project Deliverables . . . 7
- 1.6 Project Timing . . . 7

2 Assumptions and Methodology **7**
- 2.1 Systems Engineering Approach . . . 7
- 2.2 The Two Claims . . . 9
- 2.3 Proposed Airline-to-ATC Interface . . . 9
- 2.4 Experiment Design . . . 10

3 Results and Discussion **11**
- 3.1 Experiment Results . . . 12

4 Recommendations **14**

5 Impact/Financial Benefits **15**
- 5.1 Emissions: What They Mean to the Airline and to Society . . . 15

6 Summary/Conclusions **15**

A Appendices **23**
- A.1 Full Nominal-Flight ConOps . . . 23
- A.2 Full Stakeholder Register and RACCI Matrix . . . 23
- A.3 Full SysML Diagram Set . . . 23
- A.4 Full Myopic-Optimization Conflict Catalog . . . 27
- A.5 Optimization Experiment Detail . . . 27
- A.6 Generative AI Use Disclosure . . . 27
- A.7 Source Register . . . 27

1 Introduction and Current State

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Introduction. Interim Report #2 resubmits this section updated. Three edits are needed: (1) the "current state of the project" paragraphs in Section 1.2 still describe mid-September ("off-track, but recoverable", state machines due 9/15) and should describe October: the experiment defined (D-013 to D-015), the plan re-based on four tracks and nine turns (D-016), the scoring model chosen (D-017); (2) the research-gap placeholder below needs the owner's paragraph; (3) the project timing subsection should show the nine turns, not only the August table.

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

My project aims to build a reference MBSE architecture for how airspace intent is managed across the NAS — spanning the organizations that define and enforce rules, the domains managing airspace resources, the systems supporting flight operations, and the onboard systems that turn intent into executable trajectory and guidance behavior.

In his 2005 paper, DeLaurentis states the following DeLaurentis [15]:

> "Problems of significant complexity are facing decision-makers within government and industry, and the moniker of 'system-of-systems' (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the 'stovepipe' mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems ... A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time."

While this reference is from 2005, it is still relevant today. The NAS is a complex system of systems, and the decisions made by the various stakeholders have significant consequences for the safety, efficiency, and sustainability of the air transportation system.

Airline companies invest significant time and money in constructing operational plans that maximize revenue; however, on the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it Castro [10]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of experts specialized in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to minimize the impact of a disruption on the airline's schedule and revenue objectives.

The literature does not converge on a single, standardized definition of the "cost" the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. Delay cost is commonly modeled in the optimization literature as a linear function of delay minutes, but Hu et al. [42] notes that this assumption is inconsistent with airline practice, where delay cost is in fact a nonlinear function of delay time – and that, faced with this complexity, airlines tend to prioritize the speed of a recovery decision over the mathematical optimality of the resulting solution. This preference for tractable, fast decisions over globally optimal ones shows up structurally as well as behaviorally: a review of the Aircraft Recovery Problem finds that studies – and, by

extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps simultaneously, instead treating aircraft, crew, and passenger recovery as separable subproblems solved sequentially and handed off in turn Santana et al. [73]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage's constraints are applied – a concrete, literature-documented instance of the myopic-optimization pattern this capstone investigates.

That pattern's costs and benefits are visible even in work that improves on the status quo. Kontodimou et al. [54]'s predictive turnaround-buffer allocation model achieves a 37.1% reduction in propagated delay relative to a zero-buffer baseline,but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, deliberately excluding them from the optimization it performs. The gain is real within that bounded scope, but the framing illustrates the exact risk this capstone is concerned with: a locally-optimized decision layer can report a strong local benefit while externalizing tradeoffs onto adjacent decision layers that go unmeasured by that layer's own cost model. The same pattern appears on the human side of the AOCC: Turhan, Gzara, and Elhedhli [85] documents that when dispatcher workload exceeds capacity, dispatchers "may resort to conservative approaches, which can have financial implications for airlines" – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [34] frames this more broadly as a persistent gap between the disruption-management optimization literature and the reality of AOCC practice.

In this paper, I explore the value tradeoffs that aviation stakeholders experience when disruptions occur and recovery solutions are employed with various levels of myopia. When a dispatcher or AOCC team member makes a decision, they are often focused on the immediate impact of that decision – e.g., minimizing delay for a specific flight – without visibility into the broader implications for the airline's overall operations or the NAS as a whole. Cost models used today typically capture the factors that are easiest to quantify, such as fuel consumption and crew costs, while factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher/controller workload, environmental impact – are more often overlooked. This can lead to suboptimal outcomes: increased congestion in certain airspace sectors, missed opportunities for more efficient routing, or decreased overall profits driven by customer dissatisfaction. By modeling the NAS as a system of systems and analyzing the propagation of operational intent, I aim to identify strategies for improving decision-making and reducing the negative impacts of disruptions on the air transportation system.

1.1 Overview of the Project

Plan ref: to-do-list.md S1 — Describe the industry/environment (commercial aviation and the NAS), the social context, and the key stakeholders (preview – full analysis in Section 1.3). State the project's research questions from Project_To-Do List.md S1 ("Finalize project research questions") and explain why they matter. Draw on README.md's framing of the trajectory-intent lifecycle (Strategic objective -¿ mission plan -¿ flight plan -¿ ATC constraints -¿ trajectory negotiation -¿ FMS intent -¿ guidance commands -¿ aircraft motion) as the throughline.

The NAS is a complex system in which operational decisions are made by independent actors, each of which may or may not be accounting for the impacts on other system stakeholders, which includes regulatory agencies, air traffic service providers, airport operators, airline operations centers, MRO suppliers, flight crews, passengers, and aircraft systems. While substantial architectural work has been performed within individual domains, the relationships and information flows between these domains are often incompletely represented.

Existing models frequently focus myopically on one actor or problem, exhibiting one or more of the following limitations: - Emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard desision-making, - Emphasis on aircraft systems without adequate representation of the broader operational context within which those systems operate, - limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories, - difficulty evaluating the impacts of new operatioal concepts, automation technologies, and optimization services across organizational boundaries.

As a result, there is no broadly applicable architecture that fully captures how strategic objectives become executable aircraft behavior. This lack of end-to-end traceability limits the ability of engineers, operators, and decision makers to understand the consequences of changes within the NAS and complicates the integration of future decision-support technologies.

This project seeks to address this gap by developing a System-of-Systems architecture that explicitly models the propagation of intent throughout the National Airspace System, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

1.2 Review of Current State Design / Literature Review

Plan ref: to-do-list.md S2–S4 — Synthesize (do not merely list) findings from the three literature-review tracks: airline planning and operations (schedule planning, fleet assignment, crew scheduling, maintenance routing), day-of- operations and turnaround (A-CDM, weight and balance), and OCC/dispatch and flight execution (disruption management, dispatcher responsibilities, nominal ATC/IFR flow). Pull from per-source annotations captured via the literature-review workflow into `evidence/source-register.md` rather than re-deriving them here. Identify the research gap this capstone addresses (see also Project_To-Do List.md S16 "Research gap").

Plan ref: to-do-list.md Track C (research gap) — What this project adds that prior work does not. The opportunity itself is documented: [14] measured what airline-supplied weight does for climb prediction; SESAR's ADS-C EPP downlinks the FMS's predicted profile; [18] gives the aircraft a picture of the traffic. State the three things this project does that those do not: (1) scores one decision at four stakeholder levels side by side (aircraft, airline, sector, system) with no weighted sum, so who pays is visible; (2) maps the decision to its authority and its information owner in a system-of-systems model, so the missing exchange is a named interface (`AirlinePerformanceShare`, D-019) and an unmet requirement (SLR-INF-02/03/04, D-018), not a finding in prose; (3) swaps the scoring model's fidelity under a fixed architecture (D-017), which is the MBSE claim.

Plan ref: to-do-list.md S11 (context for the experiment) — What the sector controller knows when resolving a conflict. Make the point that the controller has type, level, speed and route, but not weight or cost index. Cite [75] (p. 2, weight and speed intent "are considered competitive parameters by airlines and are not available for use") and [14] (one nominal weight per aircraft type). Say who does hold the weight (airline, FMS) and that they do not hold the authority.

Plan ref: to-do-list.md S11 (context) — What controllers do by default. Cite [69] (23 of 36 level-crossing cases resolved by a descent; controllers "use more predetermined rules"), [53] (leave the aircraft in steady state alone) and [32] (instruction counts per heuristic; "No point asking aircraft to climb if they come back and say they can't"). Note the disagreement between the first and third on whether descent is preferred.

Plan ref: to-do-list.md S11 (context) — TASAR as the mirror-image study. Cite [18]. Three points to make: (1) it gives the aircraft a picture of the traffic, where this project gives ATC a picture of the aircraft; (2) size of the effect per flight (27 gallons and 2.3 minutes for the typical case, range 12 to 103 gallons, p. 12); (3) it did not model aircraft weight and limited climbs to FL350 and below for that reason (p. 8). Point (3) is the assumption Section 2.4 adopts.

Plan ref: to-do-list.md S11 (context) — How sector loading reaches the controller. Explain the Monitor Alert Parameter in one or two sentences and what a "red sector" is. Cite [26] (§18-9) for the definition and [18] (p. 10) for the practice: the sector controller is generally not aware of red sectors elsewhere, and a supervisor may restrict traffic into one. Open: the report's "area manager" is not the order's term; red versus yellow is not defined in the order.

Plan ref: to-do-list.md S11 (context) – SOURCES NOT YET HELD — Aviation emissions and what they cost whom. Needed before Sections 3 and 5 can discuss emissions: (a) the CO2 produced per kg of jet fuel, with a citation; (b) whether any carbon-pricing or offsetting scheme applies to a U.S. domestic flight, and a dated price; (c) a study of how emissions labels on flight listings affect passenger choice; (d) a social cost of carbon for the system-level discussion. None is in `evidence/sources/` as of 2026-10-09.

From my perspective, the current state of the project is off-track, but recoverable. The plan calls for 4 phases of work, with phase 1 completing 9/10 with RACII matrices for the various objectives within the airspace. More time has been spent on the literature review than originally expected, as the scope broadened to encompass consideration of which ontological model to use. Moreover, a lack of discipline on extracting specific actionable information from the review has weakened definitive claims about specific objectives pursued by stakeholders, which in turn feed the considerations of the RACII matrix. While much learning has occurred from the literature review, the impact on schedule continues. Presently, 53 papers have been extracted, with more than half manually reviewed, with many annotated. The literature review is ongoing, but the current content supports the research claims and the enforces the value of the project deliverables.

Time spent developing the Concepts of Operations has also impacted other work. However, the project lacks a specific focus at the moment for what tradeoffs will be considered. While the project does not intend to be an exploration of optimization, the concept focuses on sub-optimality driven by myopic decisions. While the concepts of operations help inform a more comprehensive understanding of the NAS, the systems involved, and the interfaces amongst them, it has not yet provided for unique insights into specific scenarios that might be worth modelling.

At this stage, the progress is at risk of analysis paralysis, which is being weighed against the risk of a lack of a well-formed vision for execution. At this point, I wish to heir on the side of action, transitioning the project to Phase 2 and allowing specific gaps or needs in the model development guide further review of the collected sources. That said, I still plan to contiue the review ad-hoc. Also worth noting is that model development will inform possibilities w.r.t. what can be modelled (e.g. multiple instances of an aircraft system performing similar, but different actions simultaneously while each affecting the same controller's workload). Additionally, gaining experience with sysmlv2 will inform whether its learning curve makes it a viable tool, or if Cameo is more appropriate.

This pivot from Phase 1 exploration to Phase 2 modelling will advance the project deliverables while engraining a deeper understanding of the information gaps. When the modelling gets to the point of MOEs/MOPs, the objectives of stakeholders will be revisited and a more targetted literature consultation can occur.

Per the original plan, state machines for the various NAS systems were intended to be completed by 9/15. This deliverable has not been met. The realities of a full-time professional job whose needs often extend beyond the normal 40-hour work week has forced much of the project work to occur on weekends and my time managment needs to adjust to accomodate. This weekend will be a decision gate that informs that process and possibly lead to an adjustment of the schedule of deliverables.

(Section 1.5).

1.3 Stakeholders

Plan ref: to-do-list.md S7 — Identify key project stakeholders (individuals/organizations affected by or perceiving themselves affected by NAS operational decisions). Draw on the PESTLE stakeholder discovery, stakeholder register, and RACCI matrix from Project_To-Do List.md S7. Confirm the stakeholder set is diverse (airline, passenger, ATC/ANSP, airport, flight crew, environmental/ societal, military – see S8's per-stakeholder objective categories) and that stakeholders with little formal power (e.g., passengers, communities affected by noise/emissions) are represented.

I created the extttcapstone extunderscore stakeholder extunderscore assessment.xlsx file, filling it with my initial assessment of each candidate stakeholder's objectives, and then iteratively refined it as I read more literature and gained a better understanding of the NAS. The assessment includes the following information for each stakeholder:

- stated interest and whether it acts as a hard constraint, optimization objective, cost/penalty, MOP, or MOE
- candidate explorations of MOP/MOEs
- determination of whether the stakeholder has a direct or indirect impact on trajectory decisions, if any.
- An exploration of whether the stakeholder should be a standalone stakeholder or consolidated with another stakeholder, and the rationale for that decision.

This has lead me to the following list of stakeholders that I believe are relevant to the project:

- Governments / Standards Organizations
- Operators (Airlines, Charter Operators, Cargo Operators)
- Air Traffic Control (ATC)
- Aircraft and Engine Manufacturers
- Airport Authority
- Flight Crew
- Passengers
- Environmental/Societal Organizations

Furthermore, an exploration of the interests of each stakeholder has lead me to the following list of objectives that I believe are relevant to the project:

- Safety / Security
- Efficiency
- Throughput / Capacity
- Reliability / Resilience
- Environmental Impact
- Noise Impact
- Passenger Satisfaction
- Schedule Reliability
- Cost / Profitability
- Maintenance / Lifecycle Cost
- Regulatory Compliance
- Workload / Human Factors

1.4 Scope

Plan ref: to-do-list.md S1 — State what is explicitly included and excluded from the System of Interest (SOI), the level(s) of abstraction modeled, and the criteria used to draw system and stakeholder/actor boundaries. This section should resolve the open boundary questions currently tracked in `knowledge/questions/open-questions.md` S1 – if any remain open when this section is drafted, that is a signal the SOI work in Project_To-Do List.md S1 is not yet closed.

See Section 1.1 for a discussion of the scope of the project. Also note ambiguities highlighted in the review of the current state design/literature review (Section 1.2) that will be addressed in future phases of the project.

1.5 Project Deliverables

Plan ref: to-do-list.md S10, S14, S16 — State the goals and deliverables: the SysML/Cameo reference architecture, the traceability chain from stakeholder needs to aircraft-level behavior, and the optimization-demonstration capability (Section 3). Include the quantifiable measures used to assess success (e.g., traceability coverage, stakeholder coverage, demonstration of a local-vs-SoS optimization tradeoff).

The following deliverables were established for the project upon its inception:

See the review of the current state design/literature review (Section 1.2) for a discussion of the current status of these deliverables and the risks to their timely completion.

1.6 Project Timing

Plan ref: to-do-list.md S1 (new) — Revisit the timeline from the 503 Proposal submittal (see `prework/ISD 503 Submittal.pdf`) and list the sequence of tasks – keyed to Project_To-Do List.md's 16 sections – with target completion dates. A "Project Timing" task did not previously exist in the plan; it has been added to Project_To-Do List.md S1 this session so this subsection has something concrete to draw from.

2 Assumptions and Methodology

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Assumptions and Methodology. This whole section is new for Interim Report #2. The minimum: (1) the working assumptions; (2) the systems engineering approach subsection (Section 2.1) with its table and the tailoring paragraph; (3) the two claims (Section 2.2); (4) the proposed interface (Section 2.3); (5) the experiment design (Section 2.4) as far as it is decided. Every `plantodo` note below names the source file for that part.

Plan ref: to-do-list.md S1 — State the working assumptions carried through the project – see README.md "Working Assumptions" and `projects/nas-sos-capstone/index.md` "Working assumptions (guardrails)" (e.g., emphasis on structure/behavior/interfaces/traceability rather than optimization math; decision-support/optimization treated as a capability inside the architecture, not the whole subject). Add the security assumption from D-019: confidentiality of shared airline data is assumed (existing SWIM and commercial software-as-a-service practice; SLR-INF-05), not modeled.

2.1 Systems Engineering Approach

Plan ref: to-do-list.md Track C (D-016) — Map the project's artifacts to the ISO/IEC/IEEE 15288 technical processes as the INCOSE Systems Engineering Handbook describes them, in one table (skeleton below), then state the tailoring in one paragraph: which process activities were reduced or omitted and why (per-paper literature checklists, full gate-to-gate ConOps, alternative decompositions, nine conflict analyses, weighted-sum optimization), citing D-016 and naming them as future work. Tailoring is a 15288 activity in its own right; say so. Cite the Handbook and the INCOSE Guide for Writing Requirements once they are registered.

Table 1: Project milestones and verification methods

Project Milestone	**Deliverable**	**Lead/Team Member**	**Expected Completion Date**	**Target Metric/Verification Method**
Literature Review	Annotated bibliography	Owen	9/7/2026	10–20 papers categorized
PESTLE Analysis & Systems-of-Interest Definition	Context diagram, text-based documentation	Owen	9/10/2026	Review
Systems-of-Interest Definition	Text-based documentation	Owen	9/15/2026	Clear boundaries and exclusions
Operational Scenarios	ConOps document	Owen	9/22/2026	1–3 representative scenarios
Draft Architecture	BDDs and package structure	Owen	10/1/2026	All domains represented
Interface Definition	Interface Control Document	Owen	10/12/2026	All interfaces identified
Internal Block Diagrams	IBDs	Owen	10/17/2026	Critical interfaces connected
Requirements Definition	Requirements model	Owen	10/20/2026	Trace to stakeholders
Measures of Effectiveness	MOE/MOP hierarchy	Owen	10/25/2026	Quantifiable metrics defined
Requirement Traceability	Requirements matrix	Owen	10/29/2026	Complete allocation
Activity Diagrams	Operational activities	Owen	11/3/2026	Scenario walkthrough
Sequence Diagrams	Information exchanges	Owen	11/8/2026	Message completeness
State Machines	State behavior of key entities	Owen	11/13/2026	Executable state logic
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed)	Owen	11/17/2026	All decisions assigned
Optimization Prototype	Working code	Owen	11/21/2026	Unit test cases
Integration into MBSE Architecture	Traceability links	Owen	11/24/2026	End-to-end scenario
Final Baseline & Final Paper	Versioned model & thesis/paper	Owen	12/16/2026	Draft
Paper Submittal	Thesis/paper	Owen	12/20/2026	Official submittal

Table 2: PLACEHOLDER – systems engineering processes and the artifacts that realize them. Wording to be the author's.

Process (15288 / INCOSE)	Artifact in this project	Where
Stakeholder needs and requirements	27 stakeholder requirements, each traced to an objective row	`requirements_stakeholders.sysml`; Appendix
System requirements	53 system requirements, baseline or improved (D-018), allocated to constituent systems	`requirements_nas_system.sysml`
Architecture definition	Context diagram, internal block diagrams, typed interfaces on the thread (D-019), activity and sequence views	`cameo_models/`; Section 3
System analysis	Experiment at three fidelities (D-017), outcome table	`simulation/`; Section 3.1
Verification	Verification cases for the improved requirements, run by the experiment	`nas_verification.sysml`
Validation	Stakeholder-level reading of the outcome table (who pays)	Section 3
Technical measurement	MoEs (outcome-table columns), MoPs (requirement measures)	Section 2.4
Configuration management	Decisions log, model text as the authoritative source (D-006), model freeze	`decisions-log.md`
Tailoring	D-016	this subsection

Plan ref: to-do-list.md S6 — Decomposition kept, alternatives named. One paragraph: why the authority / responsibility / information-ownership decomposition (D-002, D-007) was kept, naming the alternatives (organization-based, lifecycle-based, physical, information-flow, decision-authority) and the criterion (which one makes the decider, the information owner and the authority visible on the same picture). Building the alternatives is future work (D-016).

2.2 The Two Claims

Plan ref: to-do-list.md Track C (D-016, D-017) — Separate the two claims and say which the numbers support. Claim 1, tested by the experiment: replacing the decider's low-fidelity rule of thumb with a higher-fidelity model fed by weight and cost index across the airline-to-ATC interface changes which aircraft is moved and lowers total system cost (whether one extra clearance is worth it is a result, not an assumption). Claim 2, shown by the model: the architecture locates that interface, names who owns the decision and the data, states the unmet requirement, and lets the scoring model's fidelity be swapped (conceptual, nominal mass, actual mass) with the structure unchanged. The simulation would give the same numbers without the model; say so, and say what the model contributes that the numbers do not.

2.3 Proposed Airline-to-ATC Interface

Plan ref: to-do-list.md Track C (D-019) — Name the digital-thread artifact. Describe `AirlinePerformanceShare`: what it carries (gross weight, cost index, connection priority; thrust, drag and fuel-flow model identifiers present but not exercised), who sends it (airline operations center or FMS), who receives it (ATC decision support, per D-012 owned by the system that decides), and which requirements it satisfies (SLR-INF-02/03/04) and constrains (SLR-INF-05). Position it against ADS-C EPP and Coppenbarger's Table 2 (the same items, a different receiver). State that security is an assumption.

Plan ref: to-do-list.md S10 — Describe the MBSE methodology used to build the architecture in SysML v2 (textual notation authored in Syside Modeler in VS Code, with diagrams as generated views): package organization, view types produced (BDD, IBD, activity, sequence, requirements), and the traceability approach (stakeholder needs -¿ objectives -¿ requirements -¿ systems -¿ activities -¿ information exchanges -¿ aircraft behavior). Note per `workflows/update-architecture.md` (decision D-006) that the SysML v2 text is the source of truth; Cameo is not used as the source of truth. Describe the working loop (draft, print, hand review, transcribe, apply, log, re-render) and the artifact states 0 to 4 (D-016).

2.4 Experiment Design

Plan ref: to-do-list.md S11 — **State the central claim** the experiment supports. Still undecided (see `experiment-definition.md` §1 and the hand review notes 1.2, 2.1, 4.1). Answer the objection that the simulation compares two ways of operating and so does not by itself test MBSE.

Plan ref: to-do-list.md S11 — **Scenario.** Four aircraft in one invented en-route sector, shown as one snapshot: a crossing pair at the same level and two aircraft in trail 2,000 ft below. State which aircraft belong to the same airline, which is late with connections, and the weight of each. Give the level of the crossing pair and say why it is above FL350. Justify the static snapshot by the controller studies that used one ([53], [32]).

Plan ref: to-do-list.md S11 — **The one thing that differs between cases.** Define cases A, B and C by what the decider knows. State what "automated decision" means for who devises and who issues the clearance (undecided). State that the levers are the same in all three.

Table 3: PLACEHOLDER – the three cases. Wording to be the author's.

Case	Decider	Information	Climb rule applies?
A	Controller, by rule	Type, level, speed, route	Yes
B	Automation	Nominal weight per type	Yes
C	Automation	Shared weight and cost index	No

Plan ref: to-do-list.md S11 — **Altitude options and the climb rule (assumption adopted from a prior study).** State the three altitude options (2,000 ft above, 2,000 ft below, 4,000 ft below) and the rule that a climb is offered only at or below FL350, citing [18] (p. 8). Say plainly that this is that study's modeling assumption, made because it did not model aircraft weight; it is not an FAA rule and not observed controller behaviour. Then explain why the rule binds cases A and B and not case C: this is the mechanism by which missing information removes an option. Still to check: that 2,000 ft steps match the levels assigned to one direction of flight ([24], paragraph not yet found).

Plan ref: to-do-list.md S11 — **Baseline rule for case A.** One descent clearance, citing [69]. State which aircraft descends and the tie-breaker (undecided), and whether the aircraft is cleared back up.

Plan ref: to-do-list.md S11 — **Resolution options.** Introduce Table 4. For each option give the instructions it costs now, the follow-up it creates, and what the decider must know to use it. Cite [32] for the instruction accounting.

Plan ref: to-do-list.md S11 — **Sources for the option set and the counts** (owner's note, second hand review, 2.4: "Add these to report!"). Levers: `experiment-definition.md` §4, and [18] (p. 8) for lateral and altitude changes; that study does not use speed. Altitude steps and climb rule: [18] (p. 8), with the authors' reason. Instruction counts: [32] (pp. 72–74). Climbs used less than descents: [69] (6 of 36). Speed limited to ±0.04 Mach and built to fail: the author's own figure, to be stated as experience.

Plan ref: to-do-list.md S11 — **The baseline is not hobbled** (owner's note, second hand review, 3.2). In the later variation at or below FL350 the baseline may climb and does not, because its rule is one descent. Say that this matches the observed preference for descents ([69]).

Plan ref: to-do-list.md S11 — What the system column is (owner's notes, second hand review, 5.5 and 6.1). State that it is a measure of the airspace as a whole system of systems, define it by its cost statement (total fuel, total CO2, total airline cost, workload items, largest cost to any one aircraft), and refer back to that statement wherever the column is discussed.

Plan ref: to-do-list.md S11 — How each option is scored, one statement per viewpoint. Aircraft: extra fuel plus cost index times extra time ([66], Eq. 1–3). Airline: the sum over its aircraft plus a step cost when a flight misses a connection, plus CO2 converted to dollars. Sector: clearances issued, effort to devise them, and residual risk, reported as three separate items. System: totals stated side by side, with no weighted sum (D-013). Say why no weighted sum: nobody in the model owns the weights.

Plan ref: to-do-list.md S11 — Emissions. State the conversion from fuel to CO2 and its source (not yet held). State how CO2 becomes dollars at the airline level and why that route was chosen (credit price, passenger value, or both; undecided), and why the system level keeps CO2 in kg.

Plan ref: to-do-list.md S11, S12 — Truth model (decided, D-017). OpenAP (open aircraft performance model, TU Delft) scores every option: fuel flow, thrust and drag for the B737-9 at the aircraft's actual mass. State the three selectable fidelities (conceptual: feasibility and instruction counts only; nominal mass; actual mass) and that the decider's case (A to D) and the truth model's fidelity are independent settings. State the limitation: OpenAP has no drag polar for the 737 MAX 9 and borrows the MAX 8's. Cite OpenAP (Sun, Hoekstra and Ellerbroek 2020) once registered. The earlier kinematic model (`vehicle_dynamics.py`) is not used by the experiment.

Plan ref: to-do-list.md S11 — Abstractions made and why. One short table: each simplification, the reason, and the risk. A draft list is in `experiment-options-and-scoring.md` §6.

3 Results and Discussion

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Results and Discussion (preliminary). For Interim Report #2 this section needs: (1) the architecture as it stands (context diagram, aircraft IBD, the five improved requirements as the measured gap), with the figures already in Appendix A.3; (2) the two hand checks from `experiment-definition.md` §9 written up as preliminary results; (3) the first OpenAP numbers (Figure 1: fuel flow against altitude for the heavy and light aircraft) and the baseline row of Table 5 if the scoring run is done; (4) one paragraph on what is not yet done. Mark the section "preliminary" in its heading or first line, as the course asks.

Plan ref: to-do-list.md S5 — Present the nominal-flight ConOps (strategic/commercial planning through day-of-operations, turnaround, flight execution, and postflight/continuation) as the operational backbone the architecture models. Summarize; put the full activity breakdown in Appendix A.1.

Plan ref: to-do-list.md S8 — Present the stakeholder objective/cost/value ontology: what "optimal" means per stakeholder (airline, passenger, ATC/ANSP, airport, flight crew, environmental/societal, military), and the classification of each objective as hard constraint, optimization objective, cost/penalty, MOE, or MOP. Highlight conflicting vs. aligned objectives and objectives with externalized costs.

Plan ref: to-do-list.md S9 — Present the myopic-optimization analysis: the definition of "local optimum" in the SoS context, candidate SoS measures of value, and the documented conflicts (decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence). See the flagged scope note below.

Scope note (flag, not a TODO to resolve silently): Project_To-Do List.md S9 catalogs eight candidate local-vs-SoS conflicts, each with eight documented attributes (64 cells). A 15–40 page report body cannot carry all eight in full; plan to present 2–4 illustrative conflicts in the main body (favoring whichever one becomes the S11–S14 demonstration scenario) and move the remaining catalog entries to Appendix A.4.

Plan ref: to-do-list.md S10 — Present the SysML architecture: context diagram, package structure, stakeholder model, BDDs, IBDs, information-object model, interface/item-flow model, nominal-flight activity diagram, responsibility swimlanes, and requirements model with traceability. Embed the most illustrative diagrams here; move the full diagram set to Appendix A.3.

Plan ref: to-do-list.md S12, S13 — Present the optimization/simulation results: the baseline scenario, single-stakeholder vs. balanced/SoS optimization outcomes, and how simulation entities/decisions/objectives/constraints trace back to SysML blocks, properties, goals, and requirements. Show how the architecture exposes impacts a standalone optimizer would omit (S13's key point).

Scope note: Project_To-Do List.md S12 as written (Pareto fronts, full objective-weight sweeps, sensitivity analysis, tipping-point identification across the whole experiment matrix) reads as a full optimization research study layered on top of the architecture work. Per this project's own working assumption ("not an optimization paper"; optimization is a capability inside the architecture, not the whole subject) and the 15–40 page report constraint, consider scoping S12's demonstration to one representative scenario with a single weight sweep and one Pareto-style comparison, rather than the full matrix – see the note added to `knowledge/questions/open-questions.md` under "Optimization study scope (S11)."

3.1 Experiment Results

Table 4: PLACEHOLDER – resolution options. A dash means no instruction. Full version with instruction counts in Appendix A.5.

Option	AC1	AC2	AC3	AC4
O1	Descend 2,000 ft	–	–	–
O2	–	Descend 2,000 ft	–	–
O3	Climb 2,000 ft	–	–	–
O4	–	Climb 2,000 ft	–	–
O5	Turn behind AC2	–	–	–
O6	–	Turn behind AC1	–	–
O7	Reduce speed	–	–	–
O8	Descend 4,000 ft	–	–	–
O9	Descend 2,000 ft	–	Slow or turn	–

Table 5: PLACEHOLDER – outcome of each case at four levels, as a difference from case A. No simulation has been run; cells are empty.

Case	Option chosen	Aircraft (largest cost to any one)	Airline (each airline's total)	Sector (clearances / effort / risk)	System (total fuel, total CO2, total cost)
A		reference	reference	reference	reference
B					
C					

Plan ref: to-do-list.md S12 — Which option each case chose, and why. Walk Table 5 row by row. For case A, name the rule that produced the choice. For B and C, name the cost that decided it.

Plan ref: to-do-list.md S12 — Did the climb rule matter? With the crossing pair above FL350, cases A and B could not offer a climb and case C could. Say whether case C used one, for which aircraft, and what it saved. If C did not climb either, say so: that is a result too.

Plan ref: to-do-list.md S12 — B minus A, and C minus B. The first is what automation is worth; the second is what the information exchange is worth. Compare the size of each with the prior figure of tens of gallons per flight ([18], p. 12) and say whether the difference is larger than the truth model's uncertainty.

Plan ref: to-do-list.md S9, S12 — Does the best option change with the level? Read across each row of Table 5. Name any option that is better for one aircraft or one airline and worse for the sector or the system, and who pays (the largest cost to any one aircraft). This is the sub-system versus system discussion.

Plan ref: to-do-list.md S12 — Airline X's two flights. Discuss the option that touches both of one airline's aircraft: what the airline would choose between its late flight and its early one, and that the controller cannot see that preference. Connect to the dispatcher priorities (schedule integrity and connections rated at or above fuel; source of that table still to confirm).

Plan ref: to-do-list.md S12 — Workload is three things. Discuss clearances issued, effort to devise, and residual risk separately. Say where a "more workload" option on one of them is "less" on another.

Plan ref: to-do-list.md S12 — When would the controller say no? (owner's note, second hand review, 5.2). Discuss whether refusals are more likely in morning rushes at hub airports or near holidays, using the sector-loading rule in [18] (p. 10). A hypothesis; needs a source.

Plan ref: to-do-list.md S12 — Is schedule the right priority? (owner's note, second hand review, 4.11). Contrast a carbon price, which makes fuel weigh more against time, with controllers' usual practice of giving schedule the priority.

Plan ref: to-do-list.md S12 — Figure: the options on a Pareto front (owner's note, second hand review, 5.4), if that column is kept. Also needed: a figure of the scenario itself (note 1.1).

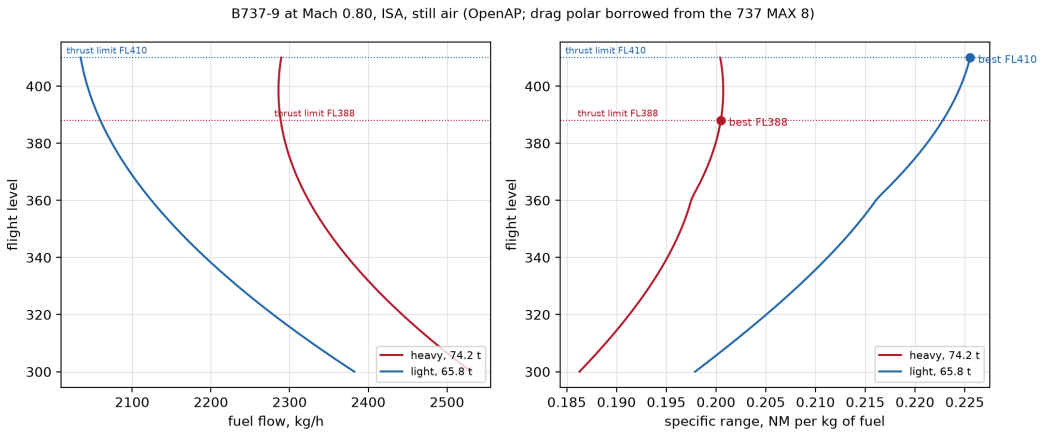


Figure 1: PLACEHOLDER – fuel flow and specific range against altitude at Mach 0.80 for the heavy and light B737-9 (OpenAP, D-017). The mechanism the experiment rests on: the light aircraft's optimum sits higher and it can climb where the heavy one cannot.

Plan ref: to-do-list.md Track A — Figure slot: sequence diagram of case C's conflict resolution (controller, ATC decision support, airline operations center, FMS, crew), rendered from the model once drafted (turn 3).

Plan ref: to-do-list.md S12 — The case where nothing changes. Report the variation with similar weights and cost indexes, where case C picks the same option as case A, to show the comparison is not rigged.

Plan ref: to-do-list.md S13 — Trace back to the model. For each column of Table 5, name the part in the context diagram it belongs to (aircraft and crew; airline operations center; ATC sector). Say what the system column maps to, since no single part owns it. Name the exchange that case C adds (weight and cost index to ATC) and where it sits in the model.

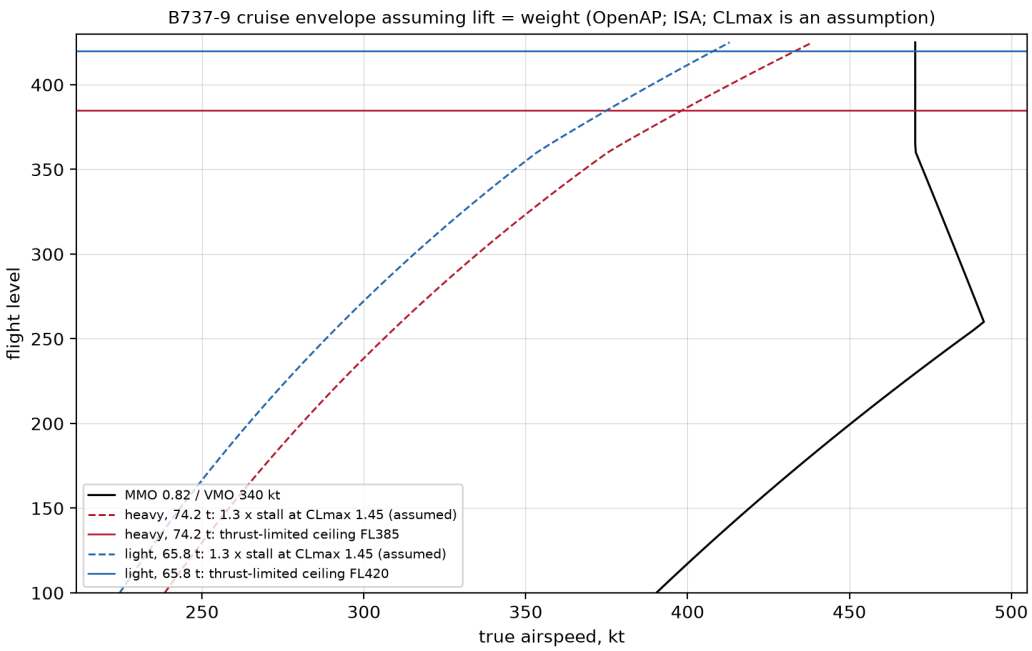


Figure 2: PLACEHOLDER – B737-9 cruise envelope assuming lift equals weight: low-speed boundary at an assumed clean maximum lift coefficient, maximum operating Mach and speed, and the thrust-limited ceiling for the heavy and light aircraft (OpenAP, D-017).

Plan ref: to-do-list.md S12, S15 — Limits. Level crossing conflicts are a minority (36 of 256, [69]); one invented sector; a static snapshot; the climb rule is one threshold for every type; airlines could misreport what they share (cite the 1 to 3 percent figure once that source is registered).

4 Recommendations

Plan ref: to-do-list.md S14, S15 (S16 bullet added this session) — Restate the main findings from Section 3 and provide recommendations for moving forward. Include the alternatives considered (e.g., alternative decompositions from S6, alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

NOTE: Project_To-Do List.md did not previously have a distinct "Recommendations" deliverable separate from "Conclusions and future work" – added to S16 this session so this section has an explicit home in the plan.

5 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its optimization-demonstration capability. Translate the S9 conflict analysis and S12 demonstration results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, gate/runway utilization) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to S8's "objectives whose costs are externalized onto another stakeholder" and S9's conflict catalog).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template's sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a "before" baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Project_To-Do List.md did not previously have a task that produces a quantified/monetized impact figure – an explicit "Impact / quantified benefits" bullet was added to S16 this session so at least one deliverable targets this section directly. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

5.1 Emissions: What They Mean to the Airline and to Society

Plan ref: to-do-list.md S9, S12 — Airline level: put CO2 into dollars. Take the fuel difference between cases from Table 5, convert to kg of CO2, then to dollars. State the route used and its source. Candidates: (1) the price of a carbon credit or allowance, if a scheme applies to the flight; (2) passenger value, meaning revenue gained or lost because some passengers choose the flight listed with lower emissions. Make the point that fuel is already counted once as fuel cost and that pricing its CO2 counts the same kilograms again, on purpose. SOURCES NOT YET HELD.

Plan ref: to-do-list.md S9, S12 — Passenger value. Discuss how passengers see a flight's emissions when booking, and whether that reaches the airline as demand. The author's own practice of choosing lower-emission listings is the motivation, not the evidence: cite a study. Note the gap between one decision in one sector and the label a passenger saw at booking; the link is the airline's average over many flights.

Plan ref: to-do-list.md S9, S12 — System-of-systems level: keep CO2 in kg. Report total CO2 across all four aircraft for each case. Discuss what it means for society and for how passengers perceive air travel. A social cost of carbon may be quoted for scale, labelled as a cost nobody in the scenario pays. Connect to S8's objectives whose costs are externalized onto another stakeholder.

Plan ref: to-do-list.md S12 — Scale it honestly. One conflict in one sector is a small number. If a per-year or fleet-wide figure is offered, state the multiplier and its source (for example, how often level crossing conflicts occur), and label it illustrative.

Plan ref: to-do-list.md S9 — Who pays. For the option the system prefers, name the aircraft or airline that is worse off and by how much. This is the equity discussion the template asks for.

6 Summary/Conclusions

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings/ recommendations, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., "reduces network-level delay propagation by an illustrative X% under the demonstrated weighting" rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the optimization demonstration (S12 scope note, Section 3), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

8.0 References

[1] Airbus Customer Services. *Flight Operations Briefing Notes: Approach Techniques: Aircraft Energy Management during Approach*. Tech. rep. FLT_OPS – APPR – SEQ03 – REV02. Airbus, Oct. 2005. URL: https://skybrary.aero/sites/default/files/bookshelf/166.pdf.

[2] Magnus Axholt and Stephen Peterson. "Modelling Traffic Scenarios for Realistic Air Traffic Control Environment Testing". MA thesis. Linköping University, 2004.

[3] Jason E. Bartolomei et al. "Engineering Systems Multiple-Domain Matrix: An Organizing Framework for Modeling Large-Scale Complex Systems". In: *Systems Engineering* 15.1 (2012), pp. 41–61. DOI: 10.1002/sys.20193.

[4] Katherine A. Berry and John J. Pace. "Examining the Actors and Functions of an Airline Operations Center". In: *Proceedings of the Human Factors and Ergonomics Society 55th Annual Meeting*. Vol. 55. 1. 2011, pp. 1412–1416. DOI: 10.1177/1071181311551294.

[5] Karl D. Bilimoria, Mike M. Paglione, and Hilda Q. Lee. "Performance Analysis of a Conflict Probe Utilizing Only State Vector Information". In: *24th International Congress of the Aeronautical Sciences (ICAS)*. Yokohama, Japan, 2004. URL: https://www.icas.org/icas_archive/ICAS2004/PAPERS/216.PDF.

[6] Katie Bolczak et al. *FAA Data Standards Initiative: Systems Engineering Base for Air Traffic Modernization*. Tech. rep. The MITRE Corporation, 2001.

[7] Soufiane Bouarfa, Jasper Müller, and Henk A. P. Blom. "Evaluation of a Multi-Agent System Approach to Airline Disruption Management". In: *Journal of Air Transport Management* 71 (2018), pp. 108–118. DOI: 10.1016/j.jairtraman.2018.05.009.

[8] Chris Brady. *Rules of Thumb*. The Boeing 737 Technical Site (web page). 2026. URL: http://www.b737.org.uk/rulesofthumb.htm.

[9] Ian Britchford. "737 MAX 8 & 9 fuel burn and operating performance". In: *Aircraft Commerce* 137 (Aug. 2021), pp. 18–26. URL: https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/General%20Articles/2021/137_FLTOPS.pdf.

[10] António Jesus Monteiro de Castro. "A Distributed Approach to Integrated and Dynamic Disruption Management in Airline Operations Control". PhD thesis. Porto, Portugal: Faculdade de Engenharia, Universidade do Porto (FEUP), 2013.

[11] Mohammad Chami et al. "Towards Solving MBSE Adoption Challenges: The D3 MBSE Adoption Toolbox". In: *28th Annual INCOSE International Symposium*. Washington, DC, 2018.

[12] Gano Chatterji. *Introduction to Air Traffic Management*. Video and transcript, Aviation Data Science Seminar (virtual). NASA Technical Reports Server (NTRS), Document ID 20205001251. NASA Ames Research Center; Universities Space Research Association, Apr. 2020. URL: https://ntrs.nasa.gov/citations/20205001251 (visited on 09/20/2026).

[13] Michael D. D. Clarke. "Irregular Airline Operations: A Review of the State-of-the-Practice in Airline Operations Control Centers". In: *Journal of Air Transport Management* 4 (1998), pp. 67–76.

[14] Richard A. Coppenbarger. "En Route Climb Trajectory Prediction Enhancement Using Airline Flight-Planning Information". In: *AIAA Guidance, Navigation, and Control Conference and Exhibit*. AIAA-99-4147. 1999, pp. 1077–1087. DOI: 10.2514/6.1999-4147.

[15] Daniel A. DeLaurentis. “Understanding Transportation as a System-of-Systems Design Problem”. In: *43rd AIAA Aerospace Sciences Meeting and Exhibit*. Reno, Nevada, 2005. DOI: 10.2514/6.2005-123.

[16] Daniel A. Delaurentis and Donald N. Fry. “Understanding the Implications for Airports of Distributed Air Transportation Using a System-of-Systems Approach”. In: *Transportation Planning and Technology* 31.1 (2008), pp. 69–92. DOI: 10.1080/03081060701835738.

[17] Ahmed E. E. Eltoukhy, Felix T. S. Chan, and S. H. Chung. “Airline Schedule Planning: A Review and Future Directions”. In: *Industrial Management & Data Systems* 117.6 (2017), pp. 1201–1243. DOI: 10.1108/IMDS-09-2016-0358.

[18] Engility Corporation. *Traffic Aware Strategic Aircrew Requests (TASAR): Annualized TASAR Benefits for Alaska Airlines Operations*. Contractor report NNL12AA06C, deliverable 41A. Corporate author, as the document states: ”prepared by Engility Corporation ... under Contract No. NNL12AA06C with NASA Langley Research Center” (Preface, p. 2); NASA Technical Monitor: David Wing (title page). No personal author is named. A web search summary of the NASA record for a later published version (NASA/CR-2015-218787, NTRS 20150017048) names Jeffrey Henderson; not confirmed and not used here. Engility Corporation, Billerica, MA, for NASA Langley Research Center, Sept. 2014. URL: https://ntrs.nasa.gov/api/citations/20140012787/downloads/20140012787.pdf.

[19] EUROCONTROL. *EUROCONTROL Specification for Airport Collaborative Decision Making (A-CDM)*. Tech. rep. Brussels, Belgium: EUROCONTROL, 2025.

[20] EUROCONTROL. *Aircraft Performance Database: BOEING 737 MAX 9 (B39M)*. Web page. 2026. URL: https://learningzone.eurocontrol.int/ilp/customs/ATCPFDB/details.aspx?ICAO=B39M.

[21] Federal Aviation Administration. *Instrument Procedures Handbook, Chapter 3: Arrivals*. Tech. rep. FAA-H-8083-16B. U.S. Department of Transportation, Federal Aviation Administration, 2017. URL: https://www.faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/instrument_procedures_handbook/FAA-H-8083-16B_Chapter_3.pdf.

[22] Federal Aviation Administration. *Flight Information Exchange Model (FIXM) US Extension v4.4.0: Logical Model Diagrams*. Tech. rep. Federal Aviation Administration / FIXM Program, 2024.

[23] Federal Aviation Administration. *Aeronautical Information Manual: Official Guide to Basic Flight Information and ATC Procedures*. Aeronautical Information Manual. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Feb. 2025. URL: https://www.faa.gov/air_traffic/publications.

[24] Federal Aviation Administration. *Air Traffic Control*. Order JO 7110.65BB. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[25] Federal Aviation Administration. *FAA Services Hierarchy*. One-page slide (PDF), FAA NAS Enterprise Architecture. Year is the PDF creation date (2025-02-24, PowerPoint export; PDF author field ”ANG-B2”); no version, title-page date or byline appears on the slide. Unverified: what ANG-B2 stands for, and whether this is the current issue of the hierarchy on the FAA NAS EA site (faa.gov/about/office_org/headquarters_offices/ang/r2r) – check before citing as current. 2025.

[26] Federal Aviation Administration. *Facility Operation and Administration*. Order JO 7210.3EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[27] Federal Aviation Administration. *Flight Services*. Order JO 7110.10EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[28] Federal Aviation Administration. *NAS Enterprise Architecture: Infrastructure Roadmaps*. Tech. rep. Federal Aviation Administration, 2025.

[29] Federal Aviation Administration. *Airport Compliance Manual, Chapter 12: Review of Aeronautical Lease Agreements*. Order 5190.6C. URL not recorded when the chapter was downloaded on 2026-09-28; add it from the FAA Order 5190.6C page before citing. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026.

[30] Federal Aviation Administration. *Airport Compliance Manual, Chapter 9: Unjust Discrimination between Aeronautical Users*. Order 5190.6C. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026. URL: https://www.faa.gov/airports/resources/publications/orders/compliance_5190_6/order-5190-6c-compliance-chapter-09 (visited on 09/28/2026).

[31] Michael Finke et al. *D2.1: Current TBO Concepts and Derivation of the Green Air Traffic Management Concepts*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2020.

[32] Selina Fothergill and Andrew Neal. "Conflict-Resolution Heuristics for En Route Air Traffic Management". In: *Proceedings of the Human Factors and Ergonomics Society 57th Annual Meeting*. 2013, pp. 71–75. DOI: 10.1177/1541931213571018.

[33] GE Aerospace. *GE Aerospace Joins MBSE Ecosystem in Michigan Aerospace*. University of Michigan College of Engineering, Aerospace Engineering news. May 2023. URL: https://aero.engin.umich.edu/2023/05/11/ge-aerospace-joins-mbse-ecosystem-in-michigan-aerospace/ (visited on 09/14/2026).

[34] L. K. Hassan, B. F. Santos, and J. Vink. "Airline Disruption Management: A Literature Review and Practical Challenges". In: *Computers & Operations Research* 127 (2021), p. 105137. DOI: 10.1016/j.cor.2020.105137.

[35] Terry R. Hill et al. "NASA's Use of MBSE and SysML Modeling to Architect the Future of Human Exploration". In: *34th Annual INCOSE International Symposium*. Dublin, Ireland, 2024.

[36] Eric C. Honour. "Systems Engineering Return on Investment". In: *INCOSE International Symposium*. Vol. 20. 1. Chicago, IL, 2010, pp. 1422–1439. DOI: 10.1002/j.2334-5837.2010.tb01150.x.

[37] Eric C. Honour. "Sizing Systems Engineering Activities to Optimize Return on Investment". In: *INCOSE International Symposium*. Vol. 21. 1. Denver, CO, 2011, pp. 1233–1245. DOI: 10.1002/j.2334-5837.2011.tb01281.x.

[38] Eric C. Honour. "Systems Engineering Return on Investment". PhD thesis. Adelaide, Australia: Defence, Systems Institute, School of Electrical, and Information Engineering, University of South Australia, 2013. URL: https://segoldmine.ppi-int.com/node/72248 (visited on 09/19/2026).

[39] Niamat Ullah Ibne Hossain et al. "Modeling and Analysis of Unmanned Aerial Vehicle System Leveraging Systems Modeling Language (SysML)". In: *Systems* 10.6 (2022), p. 264. DOI: 10.3390/systems10060264.

[40] Haoliang Hu et al. *D2.2: Report about Modelling of Operational and System Architecture*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[41] Haoliang Hu et al. *D5.1: Report on ATM Avionic System Architecture Development*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[42] Yuzhen Hu et al. "Review of Optimization Problems, Models and Methods for Airline Disruption Management from 2010 to 2024". In: *Digital Transportation and Safety* 3.4 (2024), pp. 246–263. DOI: 10.48130/dts-0024-0022.

[43] INCOSE. *Guide to Writing Requirements*. Tech. rep. INCOSE-TP-2010-006-04. Version 4. International Council on Systems Engineering, 2023.

[44] INCOSE. *INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*. 5th. Wiley, 2023. ISBN: 978-1-119-81429-0. URL: https://www.incose.org/about-systems-engineering/system-and-se-definition/systems-engineering-definition (visited on 09/19/2026).

[45] International Civil Aviation Organization. *Global Air Traffic Management Operational Concept*. Doc 9854, AN/458. Montreal, Canada: International Civil Aviation Organization, 2005.

[46] International Civil Aviation Organization. *ICAO Carbon Emissions Calculator Methodology*. Methodology document Version 13.1. International Civil Aviation Organization, Aug. 2024. URL: https://www.icao.int/sites/default/files/environmental-protection/ENVtools/SuportingMaterials/ICEC_Methodology_Passagers_v13_Final.pdf.

[47] International Civil Aviation Organization. *ICAO Aircraft Engine Emissions Databank*. Spreadsheet, hosted by the European Union Aviation Safety Agency. Issue 32. Mar. 2026. URL: https://www.easa.europa.eu/easa-and-you/environment/icao-aircraft-engine-emissions-databank.

[48] ISO/IEC/IEEE. *ISO/IEC/IEEE 15288:2023 Systems and software engineering — System life cycle processes*. Tech. rep. Geneva: International Organization for Standardization, 2023.

[49] Swapnil Sarjerao Jagtap et al. "Model-Based Systems Engineering Approach for a Systematic Design of Aircraft Engine Inlet". In: *AIAA SciTech Forum 2025*. 2025. DOI: 10.2514/6.2025-1410.

[50] Priti Jain. "Personal Knowledge Management: The Foundation of Organisational Knowledge Management". In: *South African Journal of Libraries and Information Science* 77.1 (2011), pp. 1–14. DOI: 10.7553/77-1-62.

[51] Abdessamad Jarrar and Youssef Balouki. "Formal Modeling of a Complex Adaptive Air Traffic Control System". In: *Complex Adaptive Systems Modeling* 6.1 (2018), p. 6. DOI: 10.1186/s40294-018-0056-4.

[52] Heiko Kern, Axel Hummel, and Stefan Kühne. "Towards a Comparative Analysis of Meta-Metamodels". In: *Proceedings of the Compilation of the Co-located Workshops on DSM'11, TMC'11, AGERE! 2011, AOOPES'11, NEAT'11, & VMIL'11 (SPLASH 2011 Workshops)*. ACM, 2011, pp. 7–12. DOI: 10.1145/2095050.2095053.

[53] Barry Kirwan and Mary Flynn. "Identification of Air Traffic Controller Conflict Resolution Strategies for the CORA (Conflict Resolution Assistant) Project". In: *4th USA/Europe Air Traffic Management R&D Seminar*. Santa Fe, NM, Dec. 2001.

[54] Konstantia Kontodimou et al. “Proactive Aircraft Turnaround Buffer Optimization Integrating Machine Learning and Scenario Analysis”. In: *Journal of the Air Transport Research Society* 6 (2026), p. 100117. DOI: 10.1016/j.jatrs.2026.100117.

[55] Tianning Liu et al. “Model-Based Systems Engineering Supporting Architecture Modeling of Air Traffic Management System and Model Verifying Based on SMT”. In: *Knowledge Science, Engineering and Management (KSEM 2024)*. Vol. 2269. Communications in Computer and Information Science. Springer, 2025, pp. 183–197. DOI: 10.1007/978-981-96-0178-3_13.

[56] Manoj Lohatepanont and Cynthia Barnhart. “Airline Schedule Planning: Integrated Models and Algorithms for Schedule Design and Fleet Assignment”. In: *Transportation Science* 38.1 (2004), pp. 19–32. DOI: 10.1287/trsc.1030.0026.

[57] Jing Lu et al. “Harnessing Digital Twin Technology for Enhanced Aircraft Turnaround Efficiency”. In: *Transport Economics and Management* 3 (2025), pp. 334–345. DOI: 10.1016/j.team.2025.09.001.

[58] Jinzhi Lu et al. “Design Ontology Supporting Model-based Systems-engineering Formalisms”. In: *arXiv preprint* (2020).

[59] Jinzhi Lu et al. “Semantic Model-based Systems Engineering based on KARMA: A Research and Practice Roadmap 2025”. In: *32nd Annual INCOSE International Symposium*. Detroit, MI, 2022.

[60] Christopher A. Lupp et al. “A Theoretical Foundation and Practical Demonstration of Integrating MDO, MBSE and the Digital Thread”. In: *AIAA SciTech Forum*. Air Force Research Laboratory / University of Dayton Research Institute. 2026.

[61] Apoorv Maheshwari, C. Robert Kenley, and Daniel A. DeLaurentis. “Creating Executable Agent-Based Models Using SysML”. In: *25th Annual INCOSE International Symposium (IS2015)*. Bellevue, WA, 2015.

[62] N. A. M. Maiden et al. “Model-Driven Requirements Engineering: Synchronising Models in an Air Traffic Management Case Study”. In: *16th International Conference on Advanced Information Systems Engineering (CAiSE 2004)*. Vol. 3084. Lecture Notes in Computer Science. Riga, Latvia: Springer, 2004, pp. 368–383.

[63] Tom McDermott et al. “AI4SE and SE4AI: A Research Roadmap”. In: *INSIGHT* 23.1 (2020), pp. 8–14. DOI: 10.1002/inst.12278.

[64] Faida Mhenni et al. “Flight Control System Modeling with SysML to Support Validation, Qualification and Certification”. In: *13th IFAC and IEEE Conference on Programmable Devices and Embedded Systems*. Vol. 49-3. IFAC-PapersOnLine. Elsevier, 2016, pp. 453–458. DOI: 10.1016/j.ifacol.2016.07.076.

[65] Yaniv Mordecai. “Conceptual Modeling of Cyber-Physical Gaps in Air Traffic Control”. In: *Complex Adaptive Systems Conference with Theme: Cyber Physical Systems and Deep Learning (CAS 2018)*. Vol. 140. Procedia Computer Science. Elsevier, 2018, pp. 21–28. DOI: 10.1016/j.procs.2018.10.288.

[66] Ryota Mori. “Aircraft Mass Estimation Using Cruise Flight Profile”. In: *33rd Congress of the International Council of the Aeronautical Sciences (ICAS)*. Stockholm, Sweden, 2022. URL: https://www.icas.org/icas_archive/ICAS2022/data/papers/ICAS2022_0105_paper.pdf.

[67] Sai V. Mudumba et al. “Model-Based Systems Engineering Approach for Simulating Urban Air Mobility Maturity Level 5 (UML-5) Operations”. In: *AIAA AVIATION Forum*. 2022. DOI: 10.2514/6.2022-4075.

[68] Pamela Munro and Richard Mogford. “Managing Variability: A Cognitive Ethnography of the Work of Airline Dispatchers”. In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. Vol. 62. 1. 2018, pp. 182–186. DOI: 10.1177/1541931218621043.

[69] Esa M. Rantanen and Christopher D. Wickens. “Conflict Resolution Maneuvers in Air Traffic Control: Investigation of Operational Data”. In: *The International Journal of Aviation Psychology* 22.3 (2012), pp. 266–281. DOI: 10.1080/10508414.2012.691048.

[70] Bill Roberson. “Fuel Conservation Strategies: Cost Index Explained”. In: *Boeing AERO* QTR_02 (2007), pp. 26–28. URL: https://skybrary.aero/sites/default/files/bookshelf/1956.pdf.

[71] Ítalo Romani de Oliveira et al. “A Predictive Services Architecture for Efficient Airspace Operations”. In: *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. IEEE. 2026. DOI: 10.1109/ICNS69853.2026.11570268.

[72] Ahmed R. Sadik et al. “Urban Air Mobility as a System of Systems: An LLM-Enhanced Holonic Approach”. In: *20th Annual System of Systems Engineering Conference (SOSE)*. IEEE. 2025.

[73] Mateus Santana et al. “The Aircraft Recovery Problem: A Systematic Literature Review”. In: *EURO Journal on Transportation and Logistics* 12 (2023), p. 100117. DOI: 10.1016/j.ejtl.2023.100117.

[74] Michael Schmidt. “A Review of Aircraft Turnaround Operations and Simulations”. In: *Progress in Aerospace Sciences* 92 (2017), pp. 25–38. DOI: 10.1016/j.paerosci.2017.05.002.

[75] Charles A. Schultz, David Thipphavong, and Heinz Erzberger. “Adaptive Trajectory Prediction Algorithm for Climbing Flights”. In: *AIAA Guidance, Navigation, and Control Conference*. AIAA-2012-4931. Venue and paper number unconfirmed from the PDF; DOI not found. 2012. URL: https://ntrs.nasa.gov/api/citations/20140004896/downloads/20140004896.pdf.

[76] Florian Schummer and Maximillian Hyba. “An Approach for System Analysis with Model-based Systems Engineering and Graph Data Engineering”. In: *Data-Centric Engineering* 3 (2022), e33. DOI: 10.1017/dce.2022.33.

[77] Thomas L. Seamster, Eric Chevalley, and Barbara G. Kanki. *Collaborative Systems Assessment: Flightdeck, Air Traffic Control, Flight Operations Center and Automation*. Draft Final Report, FAA task 05-04 / AJP61SSP-0050. NASA Ames Research Center, Aug. 2011.

[78] SESAR Joint Undertaking. *SESAR Solutions Catalogue 2019*. Tech. rep. Catalogue number MG-07-18-083-EN-N. Luxembourg: Publications Office of the European Union, 2019. DOI: 10.2829/998701.

[79] SESAR Joint Undertaking. *European ATM Master Plan: Making Europe the Most Efficient and Environmentally Friendly Sky to Fly in the World*. Tech. rep. SESAR Joint Undertaking, 2025.

[80] SESAR Joint Undertaking. *SESAR eATM ATM Capability Model (Extracted Capability Catalogue)*. 2025.

[81] Kapil Sheth and Sebastian Gutierrez-Nolasco. "Analysis of Factors for Incorporating User Preferences in Air Traffic Management: A System Perspective". In: *27th International Congress of the Aeronautical Sciences (ICAS 2010)*. Nice, France, 2010.

[82] Kapil S. Sheth et al. "Simulations of Credits Concept with User Input for Collaborative Air Traffic Management". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA 2010-8079. Toronto, Ontario, Canada, Aug. 2010. DOI: 10.2514/6.2010-8079.

[83] Sonali Sinha Roy et al. "From Ontology to System Architecture: An MBSE Approach Toward the Realization of Urban Air Mobility". In: *34th Congress of the International Council of the Aeronautical Sciences (ICAS)*. Florence, Italy, 2024.

[84] Junzi Sun, Jacco M. Hoekstra, and Joost Ellerbroek. "OpenAP: An Open-Source Aircraft Performance Model for Air Transportation Studies and Simulations". In: *Aerospace* 7.8 (2020), p. 104. DOI: 10.3390/aerospace7080104. URL: https://openap.dev (visited on 10/10/2026).

[85] Serkan Turhan, Fatma Gzara, and Samir Elhedhli. "Workload Balancing for Flight Dispatchers". In: *Computers & Operations Research* 186 (2026), p. 107303. DOI: 10.1016/j.cor.2025.107303.

[86] Sabine Wilke, Arnab Majumdar, and Washington Y. Ochieng. "Airport Surface Operations: A Holistic Framework for Operations Modeling and Risk Management". In: *Safety Science* 63 (2014), pp. 18–33. DOI: 10.1016/j.ssci.2013.10.015.

[87] Tim Wittenborg et al. *Knowledge-Based Aerospace Engineering: A Systematic Literature Review*. 2025.

[88] Yifan Xu, Sebastian Wandelt, and Xiaoqian Sun. "Airline Scheduling Optimization: Literature Review and a Discussion of Modelling Methodologies". In: *Intelligent Transportation Infrastructure* 2 (2024), liad026. DOI: 10.1093/itinfr/liad026.

[89] Anzhuo Yao et al. "System-of-Systems Safety for Low-Altitude Aviation Transportation". In: *Reliability Engineering and System Safety* 273 (2026), p. 112276. DOI: 10.1016/j.ress.2026.112276.

[90] Haytham Younus et al. "AI- and Ontology-Based Enhancements to FMEA for Advanced Systems Engineering: Current Developments and Future Directions". In: *Applied Sciences* 16.5 (2026), p. 2464. DOI: 10.3390/app16052464.

A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3).

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in Section 3: full BDD/IBD set, requirements model, and complete traceability matrices (needs-¿objectives-¿requirements-¿systems-¿activities-¿information exchanges-¿aircraft behavior). The figures below are every view in the model, generated by `tools/render_sysml_diagrams.py`. Not yet modeled as views: BDDs and the requirements model.

Each figure is a view defined in `nas_package_my_views.sysml` and rendered from the SysML v2 text with Syside. The caption names the view, the `.sysml` file that holds the content, and the date of the render.

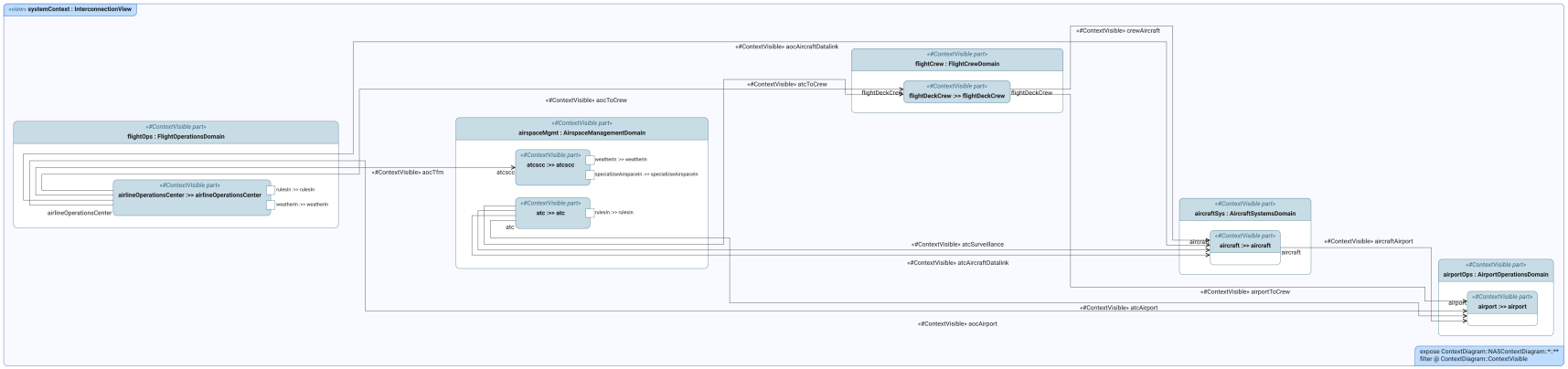


Figure 3: NAS context diagram. View `systemContext`; source `nas_context_diagram.sysml`; rendered 2026-10-10.

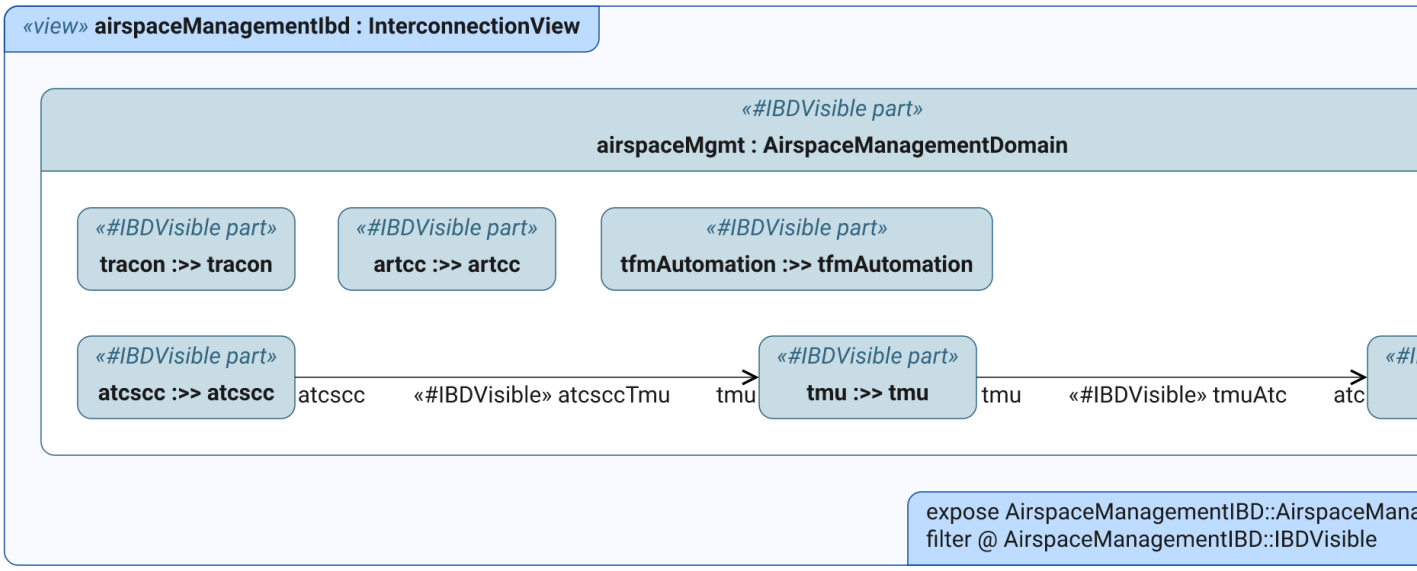


Figure 4: Airspace management internal block diagram. View `airspaceManagementIbd`; source `nas_ibd_airspac` rendered 2026-10-10.

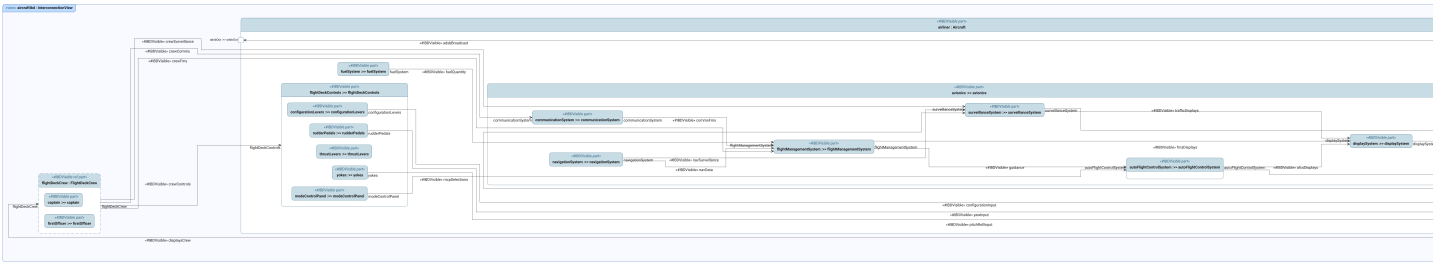


Figure 5: Aircraft internal block diagram, default level of detail. View `aircraftIbd`; source `nas_ibd_aircraft.` 10-10.

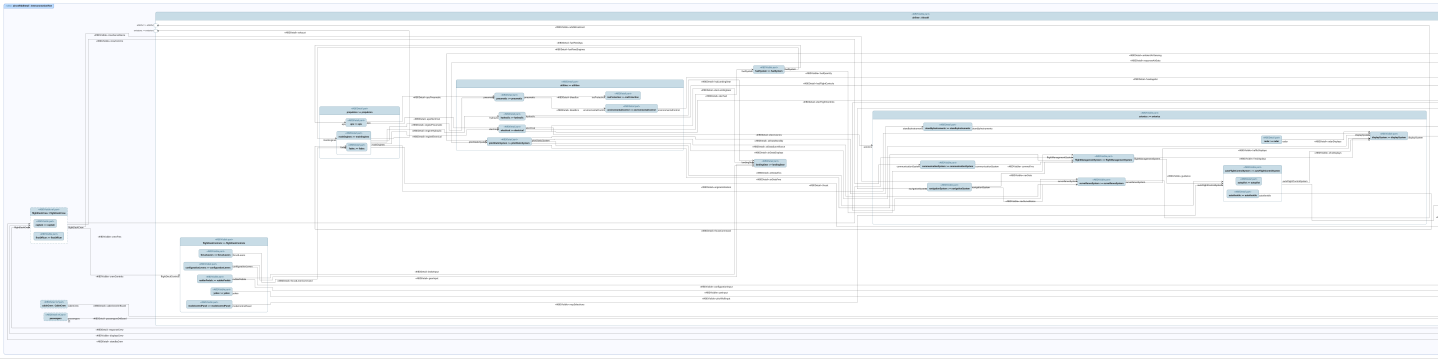


Figure 6: Aircraft internal block diagram, detailed level: adds propulsion, airframe, landing gear, utilities, and th
View `aircraftIbdDetail`; source `nas_ibd_aircraft.sysml`; rendered 2026-10-10.

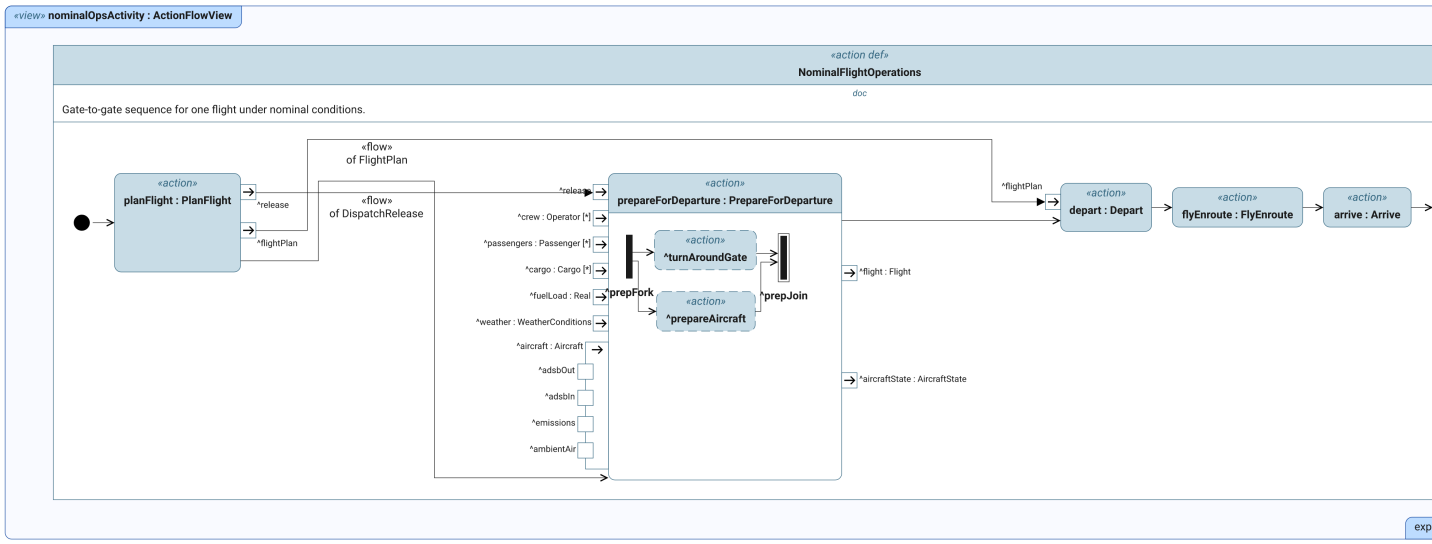


Figure 7: Nominal flight operations activity diagram (draft: no swimlanes, sub-actions are placeholders). View
source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-10.

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 3.

Plan ref: to-do-list.md S11 — Full options table: for each of the nine options, the maneuver per aircraft, instructions now, follow-up later, and what the decider must know to use it. Draft in `knowledge/models/experiment-options-and-scoring.md` §2.

Plan ref: to-do-list.md S11, S12 — Every option scored at four levels (the table from which each case picks one row), the per-viewpoint cost statements with their symbols defined, and the table of abstractions with reason and risk.

A.6 Generative AI Use Disclosure

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Generative AI disclosure: owner to draft. Required by the course policy whenever GenAI is used; Interim Report #2 is the first submission after the policy was read (2026-09-21).

A.7 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (`evidence/source-register.md`): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.